

EEID Workshop: Chronic Wasting Disease (CWD) Spatial Modeling Exercise

Background: Chronic Wasting Disease (CWD) is a fatal, contagious neurological illness affecting cervids (deer, elk, and moose). Unlike typical diseases caused by viruses or bacteria, CWD is driven by prions, which are misfolded proteins that replicate by forcing normal proteins in the host's brain to misfold as well. CWD is transmitted not only through direct animal-to-animal contact (saliva, nose-to-nose touching) but also through indirect environmental transmission. When infected deer shed prions through feces, urine, and saliva, or when an infected carcass decomposes, those prions bind tightly to clay particles in the topsoil. Once bound to the soil, prions decompose slowly and they can remain infectious in the environment for years, creating a persistent "pathogen reservoir" that healthy deer can ingest while foraging.

The Mission: Your team has been appointed as the State Epidemiological Advisory Committee. You have been given an empirical dataset built from GPS collar tracking data, which has been summarized into a spatial movement adjacency matrix (M). This matrix defines the mathematical probability of deer migrating between different grid cells across the landscape. Your objective is to evaluate how CWD spreads across different landscapes and determine which management strategies protect the state's deer herds without wasting public resources. At the end of this workshop, your team will have only 5 minutes to brief the Director of the Department of Natural Resources (DNR). You will prepare a couple of slides to present a data-driven, zone-based management plan. You must prove to the Director that your recommendations are rooted in the quantitative spatial data, and conducted simulation runs.

General presentation guidelines:

- Introduce your team and your core key question: Should the DNR manage CWD the same way across the whole state, or do we need a "Zone-Based" approach?
- Compare disease dynamics on Patchy, Corridor, and Encroachment landscapes.
- Compare different mitigation strategies and provide the DNR Director with a clear regional "Rule Book" for the state.
- Will a "one-size-fits-all" statewide regulation an epidemiological be appropriate for Chronic Wasting Disease?

The following modules will walk you through the modeling process, help you identify mitigation strategies to recommend, and prepare the information needed to make and present your slides.

Module 0: Setup and ecological concepts

Objective: To configure your R working directory, verify your empirical simulation datasets, and understand the behavioral rules governing deer movement across the state.

1. Workplace Directory & Data Verification. Before executing any code, you must ensure that RStudio can find your landscape files.

1. Create a dedicated project folder on your computer (e.g., CWD_Workshop).

2. Download and save the following four .rds landscape files directly into that project folder:
 - *cwd_data_random.rds*
 - *cwd_data_patchy.rds*
 - *cwd_data_corridor.rds*
 - *cwd_data_encroachment.rds*
3. Open *Run_CWDSimulation.R* R script in RStudio.
4. Set your working directory to your project folder by navigating to Session > Set Working Directory > To Source File Location (or by manually running `setwd()`).

2. **Observe the outputs:** Running *Run_CWDSimulation.R* should generate two kinds of outputs:
- RStudio Plot Pane: The console immediately renders a couple of plots
 - Local Directory Video Export: The script makes a .mp4 video file saved directly to the user's active working directory.

3. Understanding the Movement Matrix (M): To simulate realistic spatial dynamics, this workshop uses a Eulerian framework to model the continuous "flow" of host population densities (S and I) and pathogen loads (P) across a structured grid. The movement of deer between cells is governed by an Adjacency Matrix (M) derived from synthetic GPS collar telemetry data. The modeling code shifts a deer's behavior dynamically based on three specific habitat types, using the following baseline probabilities to dictate how long a herd stays within a given grid cell:

- **Forest Patches:** Represent baseline protective cover. Deer have a 90% stay probability here.
- **Agricultural Fields:** Act as high-quality forage patches. The code assumes a high 96% stay probability, meaning deer linger extensively in these feeding zones.
- **Suburban Patches:** Act as lower-quality or high-risk fragments due to human development. The code assigns a low 40% stay probability here.

Module 1: Landscape comparison

Objective: To observe how the spatial arrangement of habitats alters the transmission of Chronic Wasting Disease and the formation of environmental prion reservoirs.

Instructions:

1. Run your base R script *Run_CWDSimulation.R* using different input files in Section 1. Between these runs look at the plots created and the .mp4 file saved in your working directory. (Note, you might want to save your figures and rename the saved *CWD_Simulation_Output.mp4* file between runs so it doesn't get written over.)
 - *cwd_data_random.rds* (Control)
 - *cwd_data_patchy.rds* (Fragmented high-density patches)
 - *cwd_data_corridor.rds* (Linear travel networks)
 - *cwd_data_encroachment.rds* (Urban Fragmentation / Subdivided Wildlife Refuges)
2. Watch the resulting .mp4 video for each and analyze your snapshots plots.

Data Collection & Analysis Questions

1. For each landscape, look at the bottom-left plot (Herd Prevalence). At what approximate time step does the disease prevalence reach its absolute peak?
 - Random landscape: T = _____
 - Patchy landscape: T = _____
 - Corridor landscape: T = _____
 - Encroachment landscape: T = _____
2. Why did the disease move at such different speeds? Look at how the deer are distributed spatially on the top-left map to justify your answer.
3. Look closely at the bottom-right plot (Total Prion Accumulation) alongside the Prevalence curve. When the infected host population plummets or burns out, does the global prion load instantly plummet too? Describe the shape of the purple line at the very end of the simulation (T = 150 to T = 200).
4. Look at the final frame (T=200) on the Prion Soil Map (Top Right) across all four landscapes. Even if the current number of infected deer is very low, what does the remaining purple spatial pattern tell you about the future safety of reintroducing healthy deer into those specific patches?

Module 2: Parameter manipulation for simplified mitigation

Objective: To evaluate standard wildlife management regulations by altering structural biological parameter values without changing code architecture.

Instructions

1. Open you're the *Run_CWDSimulation.R* script and locate Section 4 (BIOLOGICAL PARAMETERS).
2. Modify the assigned values below one scenario at a time, run the script on *cwd_data_encroachment.rds*, and analyze the resulting dashboard.
3. **Important:** Reset all values back to their defaults before starting the next scenario!

Scenario 1: Feeding and Baiting Bans: Wildlife agencies often ban hunters from leaving out bait piles or landowners from putting out feed troughs, as these draw deer into artificial, face-to-face crowds. Lower direct transmission (β_d) from 0.012 to 0.004. Leave all other parameters at defaults.

1. Look at the bottom-left plot (Prevalence). How does reducing direct nose-to-nose contact alter the overall peak height and the timeline speed of the epidemic?
 - a.) Peak Height: _____
 - b.) Timeline Speed: _____
2. Check the purple line on the bottom-right plot (Total Prion Accumulation). Did this management policy stop the long-term contamination of the soil reservoir, or did it merely delay the accumulation curve? Look at the values at $T = 100$ and explain the biological mechanism why.

Scenario 2: Soil Remediation Treatments Imagine a future management breakthrough where fields are treated with an enzyme spray that aggressively breaks down environmental prions bound to plants and topsoil. Reset β_d to 0.012. Increase environmental decay (γ) from 0.01 to 0.08.

1. How did increasing the environmental decay rate alter the visual appearance of the Prions in Soil Map (Top Right) at $T=200$? Do you see the same lingering "ghost trails" of purple that you saw in the baseline model?
2. Look at the Total Prion Accumulation plot (Bottom Right) for this run. What is the approximate maximum value reached by the purple line, and how does this compare to the baseline model?

Module 3: Programing interventions for more complex mitigation

Objective: To implement active, automated management responses by programming conditional logic directly into the simulation loop.

PART A: Guided Coding (Copy, Paste, & Analyze): Your objective is to deploy a culling program that directly targets and removes infected individuals. Once the total number of infected deer tracked across the entire state hits an alert threshold of 100 individuals, agency field teams are deployed to active infection hotspots. This intervention aims to remove 50% of the infected deer (I). However, because field operations are imperfect, the mathematical assumption of this program is that an incidental 15% loss of susceptible deer (S) is incurred during the intensive culling process

Instructions

1. Locate the # STUDENT EXERCISE block inside the main simulation loop (Section 6).
2. Paste the following conditional block exactly into that space:

R code:

```
# Policy 1: Global Emergency Density Reduction
global_trigger <- 100

if (sum(I) >= global_trigger) {
  S <- S * 0.85 # 15% incidental loss of healthy deer during
  culling
  I <- I * 0.50 # 50% targeted removal of infected deer
}
```

3. Save, run the simulation on `cwd_data_random.rds`, and answer the following:

Questions

1. Run your updated script with the culling logic on the `cwd_data_random.rds` landscape. Look at the dark red line on the bottom-left plot (Prevalence). Describe the unique, jagged visual pattern that appears over time. What is happening biologically at each sharp downward drop and subsequent rise?
2. Look at the end of the timeline (T=200) for Policy 1. Compare your final surviving healthy population (S = 607) against the baseline Random model simulation where no culling took place (S = 1985). Did deploying this aggressive culling program end up saving more deer globally across the landscape? Explain why or why not.
3. Check the bottom-right plot (Total Prion Accumulation). How did this continuous culling strategy impact the long-term contamination of the soil matrix compared to the baseline model?

PART B: Independent Coding Challenge (Write Your Own Policy!): Now it is your turn to design a policy from scratch. Biologists know that disease transmission drops during severe seasonal lockdowns or when high-risk boundaries are physically fenced off.

The Scenario: Your agency decides to pass an emergency quarantine order. To make a real impact before the environmental reservoir becomes too heavily contaminated, they want you to deploy an aggressive intervention that slashes both direct contact transmission (β_d) and indirect soil transmission (β_p) by 80%, ONLY after time step 50 ($T > 50$).

Instructions:

1. Do not overwrite your culling code! In RStudio, go to File -> Save As... and name this new script file *Run_MitigationQuarantine.R*, then erase the culling code from Part A out of the # STUDENT EXERCISE placeholder box.
2. Write a custom if statement that checks if the current time step t is greater than 80.
3. Inside that statement, redefine β_d and β_p to be half of their original values.
4. *Hint:* Your code structure should mimic this format:

R code:

```
# Policy 2: Mid-Epidemic Quarantine
if (t > 50) {
  # Write lines here to modify beta_d and beta_p
}
```

Questions:

Run your new culling and quarantine scripts across the three landscape files. Complete the matrix below for the DNR Director. Instead of a single rating, evaluate how well each policy achieves its specific target by writing **Highly effective**, **Moderately effective**, or **Ineffective** for both output measures:

- Herd Preservation: Did it save living, healthy deer (S values in the title)?
- Environmental Protection: Did it keep the topsoil clean (Purple lines near zero)?

Mitigation Strategy	Output Measure	Patchy	Corridor	Encroachment
Reactive local culling (Part A)	Herd Preservation			
	Environmental Protection			
Delayed Quarantine (Part B)	Herd Preservation			
	Environmental Protection			

2. Delayed Quarantine (Part B) wins for immediate hunter satisfaction, while Reactive Local Culling (Part A) wins for long-term environmental protection. Culling targets the host reservoir.

It keeps the environment pristine because it completely eliminates shedding vectors, but it inflicts brutal collateral damage that decimates living herds. Quarantine targets the transmission pathway. It keeps the living herd thriving (S values increase relative to baseline), but because infected hosts remain on the map, they continue to slowly drop prions, allowing the soil reservoir to grow on crowded or heavily networked landscapes.

3. Based on how these strategies perform across the different maps, would you recommend a single, "one-size-fits-all" statewide regulation policy for this disease? Why or why not?